

What contributes to the success of IT projects?

Success factors, challenges and lessons learned from an empirical study of software projects in the Norwegian public sector

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Abstract—Context. Each year the public sector invests large amounts of money in the development and modifications of their software systems. These investments are not always successful and many public sector software projects fail to deliver the expected benefits. **Goal.** This study aims at reducing the waste of resources on failed software projects through better understanding of the success factors and challenges. **Method.** Thirty-five completed software projects in 11 organizations in the public sector of Norway were analyzed. For each project, representatives from the project owners, project management and the user organization were interviewed. **Results.** Small and large software projects reported different challenges, especially related to project priority. Taking advantage of agile practices such as flexible scope and frequent delivery increased the success rate of the projects. Projects with time and material contracts and involved clients during execution were more successful than other projects. The respondents experienced that extensive involvement and good competence of the client, high priority of the project, good dialogue between client and provider and appliance of agile practices were main success factors. Main challenges were related to technical issues, project planning and management, transition of the product to the user organization, involvement and competence of the client, and benefit management. **Conclusions.** Success factors tend to focus on human factors, e.g., involvement, competence and collaboration. Challenges focus on human factors as well as issues of technical nature. Both aspects need to be addressed to enable successful and avoid failed software projects. Competence, client involvement and benefit management are among factors that the public sector should focus on for realizing client benefits.

Keywords—software projects; success; empirical study; agile; contract type

I. INTRODUCTION

Over decades, organizations have invested large sums on Information Technology (IT) and the budgets continue to grow. Naturally, they are interested in knowing the return on these investments. Recent surveys suggest that as much as 10-15% of all software projects are stopped before they delivered anything at all [4, 5]. The importance of improving software development performance, including avoiding failures, has led to numerous surveys on failure factors of

software projects, such as those reported in [1, 2, 3, 4, 7, 8, 9, 10]. In spite of extensive opportunities to learn from experience, it seems that software projects fail for very much the same reasons as before.

The SMIOS research project, led by the Simula Research Laboratory, started in April 2015 and includes 11 partner organizations in the public sector of Norway. SMIOS is the Norwegian acronym for «Success with IT-Projects in the Public Sector». The SMIOS project aims at generating and disseminating knowledge about how to increase the benefit and success rate of investments in IT-development, based on the experiences of partner organizations. The results reported in this extended abstract are from the first phase of the SMIOS project, where information about 35 completed software projects were analyzed. The focus of the study is on the client perspective of project success or failure.

II. STUDY DESIGN AND OVERVIEW OF THE PROJECTS

We asked each of the 11 partners of the SMIOS project to select their 2-4 last completed or stopped software projects in 2014. An emphasis on the last projects was aimed to avoid a self-selection bias. A few of the organizations did not use only the last projects, but selected projects they considered as good examples of successful and not so successful projects. This deviation is not likely to significantly affect the results of the study.

For each project, we performed semi-structured interviews with representatives from the project owners, project management and the user organization. In total 107 interviews were conducted. We asked about the project characteristics, applied processes for development as well as project management, resource management and benefit management, project outcomes (in terms of cost, quality, delivered functionality and benefits), and perceived success factors, challenges and lessons learned. Additionally, we read available project documentation. For the purpose of our analysis, we defined a *successful project* as (also see [6]):

A successful project is a project that is assessed by the interviewees to deliver the expected client benefits or more, and where none of the respondents report large or very large problems on the success dimensions time control, budget control, quality and delivered functionality.

Admittedly, the definition of success is subjective. Interestingly, the respondents in the different roles had very

similar judgment of the degree of success and problems in the project.

Some characteristics of the analyzed projects are:

- The planned budget varied between 10 000 Euros and 100 million Euros. The median budget was around four million Euros.
- Types of contracts used with external providers were *fixed price* (4), *time and material* (12), *risk sharing/target price* (16) and *other* (3). Some risk sharing contracts had lower and upper limits for risk sharing (9) while others did not have such limits (7).
- All but two projects used incremental or agile development methods. The frequency of delivery to *production* varied from one delivery (13), to several deliveries (two to four) (13), and frequent (more than four) deliveries (9) during the project execution.

III. SOME INTERESTING FINDINGS

Using the definition of success and failure given before, 25 of the projects were categorized as successful, 5 as problematic and 5 as very problematic (including 2 projects that were stopped before completion). The project success rate of our sample is consequently approximately 70%.

We did not find any strong connection between project size and its success. Of the 10 projects that were not successful, three were categorized as large, two as medium large and five as small. Our results however suggest that larger and smaller project have different challenges, e.g., related to priority, client involvement and access to best competence.

The relation between the success rate of the projects with agile development and scope flexibility is:

- Projects with agile development and fixed scope had a success rate of 58% (12 projects).
- Projects with agile development and flexible scope had a success rate of 87% (15 projects).

Additionally, we realized that frequent delivery increases the success rate of projects. In fact, all (8) projects with flexible scope and frequent delivery were categorized as successful. The respondents were clear in their opinion about how applying these agile practices allowed changing the scope based on external changes, learning through getting high quality feedback from representative users, and achieving more client benefit.

Fixed price projects in our sample were more problematic than other projects:

- Projects with fixed price contracts: 38% successful
- Project with time & material contracts: 83% successful

Risk sharing contracts with an upper limit for payment may be experienced as fixed price (and financial loss for the provider) in situations with high cost overrun. We observed that clients using fixed price type of contracts tend to have a stronger focus on low price and a lower focus on evaluating provider competence when selecting providers (*fixed price behavior*). Additionally, projects with fixed price type of contracts are less likely to deliver frequently and implement benefit management during the project execution, which are

practices that are mentioned by the clients as important to the success of the projects.

In total, we collected over 200 responses related to success factors, 120 responses related to challenges and over 100 responses related to what should be done differently in a similar situation (lessons learned). The responses were coded using categories, which emerged from repeated reading and analysis of the responses.

For success factors, 26% of responses were related to the involvement of stakeholders with high priority and good dialog between the client and provider most frequently mentioned. *Competent client* was mentioned in 10% of the responses while *competent provider* was mentioned in 13% of them. The usage of agile practices had the highest frequency among the factors related to the development process (9%).

Involving stakeholders and competence were most frequently mentioned challenges. Additionally, planning and initiation phase were perceived as challenging for public sector projects, especially for larger projects (19%); for example poor estimation process, unrealistic scope and unclear requirements. Technical issues (architecture choices, integration, integration with Commercial-Off-The-Shelf products and testing) were perceived by the clients as challenging as well (7%). Finally, benefit management is an area of concern with 7% of responses.

Not surprisingly, lessons learned highlighted success factors as well as suggested solutions to overcome the challenges. The interviewees specially focused on early delivery that allows learning as well as realizing benefits, involvement of the stakeholders (including involving the user organization and planning for the transition), right competence and better processes for managing resources, and improving the project management practices.

Our sample of projects is not large enough to make strong claims and probably some results are biased towards public organizations with better than average competence. Nevertheless, the study highlights factors to be extra aware of as a client in the public sector.

IV. CONCLUSIONS

Some findings of the SMIOS project are presented in this extended abstract while we plan to publish the detailed results in future. The results, we hope, contributes to the body of empirical knowledge on the characteristics of successful software projects and typical challenges. Based on our experience from software projects outside Norway, and the similarity in success and failure factors reported across different nations' software projects, we believe that the value of the findings will not be limited to Norwegian software projects.

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REFERENCES

- [1] M. Bloch, S. Blumberg and J. Laartz, "Delivering large-scale IT projects on time, on budget, and on value", *Harvard Business Review*, June 2012.
- [2] T. Chow and D.-B. Cao, "A survey study of critical success factors in agile software projects", *Journal of Systems and Software*, **81**(6), 2008, pp. 961-971.
- [3] A. Cole, "Runaway projects: cause and effects", *Software World* **26**(3), 1995, pp. 3-5.
- [4] K. El Emam and A. G. Koru, "A replicated survey of IT software project failures", *IEEE Software* **25**(5), 2008, pp. 84-90.
- [5] M. Jørgensen, "Failure factors of small software projects at a global outsourcing marketplace", *Journal of Systems and Software* **92**, 2014, pp. 157-169.
- [6] M. Jørgensen, "A survey on the characteristics of projects with success in delivering client benefits", *Information and Software Technology*, **78**, 2016, pp. 83-94.
- [7] L. A. Kappelman, R. McKeeman and L. Zhang, "Early warning signs of IT project failure: The dominant dozen", *Information Systems Management* **23**(4), 2006, pp. 31 - 36.
- [8] L. McLeod and S. G. MacDonell, "Factors that affect software systems development project outcomes: A survey of research", *ACM Computing Surveys (CSUR)* **43**(4), 2011, article no. 24.
- [9] R. Schmidt, K. Lyytinen, M. Keil and P. Cule, "Identifying software project risks: An international delphi study", *Journal of Management Information Systems* **17**(4), 2001, pp. 5-36.
- [10] K. T. Yeo, "Critical failure factors in information system projects", *International Journal of Project Management* **20**(3), 2002, pp. 241-246.